Biology practice paper 2: answers and marks

How to mark your paper. Go through one question at a time. Read the answer, then check your own answer against "How the marks are given". Give yourself the mark if your answer means the same thing, even if you used different words.

Write your mark next to each question, then add them up. Read the "Watch out" boxes even when you got the marks: they're the mistakes that cost people marks in the real exam.

01.1

1 mark

Which part of a cell controls its activities? Tick one box.

The answer

Nucleus.

How the marks are given

- 1 mark: nucleus.

Watch out

- The cell membrane controls what goes in and out, not the cell's activities.

01.2

1 mark

Bacterial cells are prokaryotic. Give one way a bacterial cell is different from an animal cell.

The answer

A bacterial cell has no nucleus: its DNA is in a single loop in the cytoplasm.

How the marks are given

- 1 mark for any one: no nucleus; DNA in a loop (or free in the cytoplasm); has plasmids; has a cell wall; much smaller.

Watch out

- Saying "it's smaller" is fine, but "it has DNA" isn't a difference: both have DNA.

01.3

2 marks

A cell is 0.02 mm wide. An image of the cell is 4 mm wide. Calculate the magnification. Use: magnification = size of image ÷ size of real object. Show your working.

The answer

×200. Working: 4 ÷ 0.02 = 200.

How the marks are given

- 1 mark: the working, 4 ÷ 0.02.
- 1 mark: the answer, 200 (or ×200).

- The question asks you to show your working, so the answer on its own only gets 1 mark.

Watch out
- Image size goes on top. 0.02 ÷ 4 gives 0.005, which is wrong.

01.4

2 marks

Describe what osmosis is.

The answer
Osmosis is the movement of water from a dilute solution to a more concentrated solution, through a partially permeable membrane.

How the marks are given
- 1 mark: you say it's the movement of water.
- 1 mark: from a dilute solution to a more concentrated one, through a partially permeable membrane.

Watch out
- It's water that moves, not sugar or salt.
- Learn it word for word: water moves from dilute to concentrated.

01.5

1 mark

A plant cell is put into a very concentrated salt solution. What happens to the cell? Tick one box.

The answer
It loses water and shrinks.

How the marks are given
- 1 mark: loses water and shrinks.

Watch out
- The salt solution is more concentrated than the cell, so water moves out of the cell by osmosis. A plant cell can't burst, because its cell wall stops it.

02.1

1 mark

Which enzyme breaks down protein? Tick one box.

The answer
Protease.

How the marks are given
- 1 mark: protease.

Watch out

- Bile isn't an enzyme: it breaks fat into droplets. Amylase breaks down starch and lipase breaks down lipids.

02.2

1 mark

Name the products of protein digestion.

The answer

Amino acids.

How the marks are given

- 1 mark: amino acids.

Watch out

- Fatty acids and glycerol come from lipids, not proteins.

02.3

2 marks

A student tested a food. The table shows the results. Iodine solution: Blue-black; Benedict's solution (heated): Blue; Biuret reagent: Purple.

Which two types of food molecule does the food contain? Use the results in the table.

The answer

Starch and protein.

How the marks are given

- 1 mark: starch (iodine went blue-black).
- 1 mark: protein (Biuret went purple).

Watch out

- Benedict's stayed blue, so there's no sugar. Benedict's would have turned brick red if there was.
- Iodine: orange-brown to blue-black for starch. Benedict's: blue to brick red for sugar. Biuret: blue to purple for protein.

02.4

2 marks

Explain why amylase can break down starch but cannot break down protein.

The answer

Amylase has an active site with a particular shape. Only starch fits into it, so protein can't be broken down by amylase.

How the marks are given

- 1 mark: you say the active site has a particular (specific) shape.
- 1 mark: you say only starch fits the active site (protein doesn't fit).

Watch out

- This is the lock and key idea: each enzyme fits only one substrate.
- Never say an enzyme is "killed". Enzymes aren't alive.

02.5

6 marks

A student wants to find out how temperature affects how quickly amylase breaks down starch. Describe a method the student could use. Include how they would make it a fair test.

A full-marks answer

Put some starch solution and some amylase solution in separate test tubes in a water bath at 20 °C, and leave them for a few minutes to reach that temperature. Put a drop of iodine in each well of a spotting tile. Mix the amylase and starch and start a stopwatch. Every 30 seconds, put a drop of the mixture into a new well of iodine. When the iodine stays orange-brown, the starch has all gone: write down the time. Do it again at 30 °C, 40 °C, 50 °C and 60 °C. To make it a fair test, use the same volume and concentration of starch and amylase each time, and keep the pH the same with a buffer solution. Repeat each temperature and work out a mean.

How the marks are given

This question is marked as a whole. Read your answer, then pick the description that fits it best.

5 to 6 marks	Your method is clear, in the right order, and would work. You say how you change the temperature (water baths), how you know the starch has gone (iodine stays orange-brown) and how you time it, at least two things you keep the same, and that you repeat it.
3 to 4 marks	Your method would mostly work, but a step is missing or unclear. You change the temperature and test for starch, and keep at least one thing the same.
1 to 2 marks	You give some steps, but not in a clear order, or you don't say how you'd know when the starch has gone.
0 marks	Nothing that would help find the answer.

Check you included

- Water baths at different temperatures, such as 20, 30, 40, 50 and 60 °C
- Let the solutions reach the temperature before mixing
- Mix the amylase and starch and start a stopwatch
- Test a drop with iodine every 30 seconds on a spotting tile
- Stop the clock when the iodine stays orange-brown
- Keep the same: volume and concentration of starch and amylase, pH (buffer), time between tests
- Repeat and work out a mean

03.1

1 mark

Salmonella is a pathogen that causes food poisoning. What type of pathogen is it? Tick one box.

The answer
Bacterium.

How the marks are given

- 1 mark: bacterium.

Watch out

- Measles and HIV are viruses. Rose black spot is a fungus. Malaria is a protist.

03.2

2 marks

Give two ways the body stops pathogens getting in.

The answer
The skin acts as a barrier, and the stomach makes acid that kills pathogens.

How the marks are given

- 1 mark for each, up to 2 marks: skin; hairs and mucus in the nose; cilia and mucus in the trachea and bronchi; stomach acid.

Watch out

- White blood cells fight pathogens once they're already inside. This question is about stopping them getting in.

03.3

2 marks

A doctor gives a patient antibiotics for a bacterial infection. Explain why the doctor would not give antibiotics to someone with a cold.

The answer
Antibiotics kill bacteria, but a cold is caused by a virus. Antibiotics don't work on viruses.

How the marks are given

- 1 mark: you say a cold is caused by a virus.
- 1 mark: you say antibiotics don't kill viruses (they only kill bacteria).

Watch out

- "The cold is too strong" or "the virus is resistant" are wrong. Antibiotics never work on any virus.

03.4

1 mark

What do painkillers do? Tick one box.

The answer
They relieve the symptoms.

How the marks are given

- 1 mark: they relieve the symptoms.

Watch out

- Painkillers don't kill pathogens or cure anything. They just make you feel better while your body fights the disease.

04.1

1 mark

Complete the word equation for photosynthesis: carbon dioxide + water → _____ + oxygen

The answer

Glucose.

How the marks are given

- 1 mark: glucose.

Watch out

- Oxygen is already in the equation. Photosynthesis needs light, but light isn't a product.

04.2

2 marks

Give two factors that affect the rate of photosynthesis.

The answer

Light intensity and carbon dioxide concentration.

How the marks are given

- 1 mark for each, up to 2 marks: light intensity; carbon dioxide concentration; temperature; amount of chlorophyll.

Watch out

- "Light" on its own is fine, but "sun" or "weather" isn't specific enough.

04.3

1 mark

During hard exercise, muscles can respire anaerobically. Name the substance made by anaerobic respiration in muscles.

The answer

Lactic acid.

How the marks are given

- 1 mark: lactic acid.

Watch out

- Ethanol and carbon dioxide come from anaerobic respiration in yeast, not in muscles.

04.4 **1 mark**

Give one reason why aerobic respiration is better for the body than anaerobic respiration.

The answer

Aerobic respiration releases much more energy.

How the marks are given

- 1 mark for any one: it releases more energy; it doesn't make lactic acid (no oxygen debt or muscle fatigue).

Watch out

- Respiration releases energy. Never say it "makes" or "creates" energy.

Total: 30 marks. A rough guide for this practice paper: 22 or more is a strong result. 15 to 21 is solid, with some gaps to fix. Under 15 means going back over these topics in the notebook first. Whatever you got, write down the questions where you lost marks: those are what to revise next.